

Project Design Phase Solution Architecture

Date	03 Oct. 26
Team ID	SWTID-2026-8365
Project Name	Import Data using Transform Maps (Spreadsheet) – ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

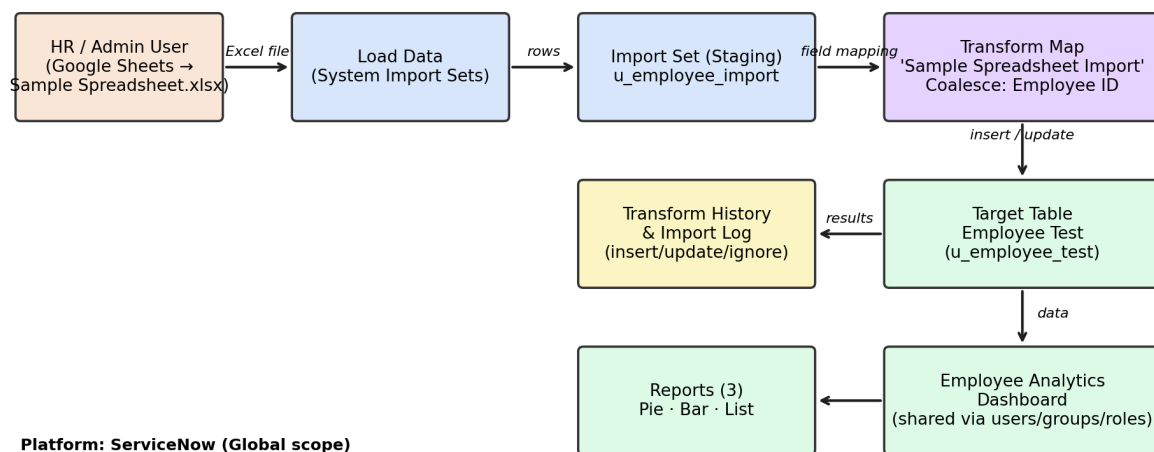


Figure 1: Architecture and data flow of the employee data import, transformation and reporting solution on ServiceNow

Components: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

- Source: Employee data (Employee ID, Name, Email, Department, Location) prepared in Google Sheets and downloaded as Sample Spreadsheet.xlsx.

- Load Data: loads the Excel file (sheet 1, header row 1) into the staging Import Set table u_employee_import.
- Transform Map: maps source fields to the target table u_employee_test (Employee Test); Coalesce on Employee ID updates existing records and inserts new ones.
- Validation: Transform History and Import Log show total, inserted, updated, ignored and error counts.
- Reporting: three reports (Employees by Department – pie, Employees by Location – bar, Employee List – list) are added to the Employee Analytics Dashboard, which can be shared with selected users, groups and roles.